Attentional Control Theory

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Learning outcomes, Session I

From working memory and attention, through what anxiety does to them, to the quiet eye. By the end of this session you should be able to:

1. Describe attentional control, in terms of goal-directed and stimulus-driven orienting
2. Define cognitive anxiety (worry) and distinguish it from the other forms of anxiety
3. Explain the distinction between processing efficiency and performance effectiveness
4. Describe empirical evidence that anxiety biases attention towards threat
5. Define the quiet eye, as a variable and as a phenomenon
6. Recognise when a performance problem can be due to issues with attentional control

Optional material

> **Note**
>
> Content in slides marked **Optional** adds depth. This content is not examined in the multiple-choice exam. However, it can support a stronger, more critical essay assessment. Press the down arrow / scroll down where you see the cue.

Attentional Control Theory, in a nutshell

Anxiety consumes working memory and transfers attention from goal-directed to stimulus-driven control ; performers compensate with extra effort, so efficiency drops before performance does.

In lay terms:

1. Worry takes resources. There is less left for the task.

2. Attention is called to whatever is salient rather than by what the task requires.

3. You can maintain performance by trying harder but you would be using more resources = less efficient.

Agenda

Route through Session I: working memory and attention → what anxiety does → processing efficiency theory → the quiet eye → signs to look for

Worksheet 1

Worksheet 1

Space to write during this session: predictions, reasoning, notes.

Open it now and keep it alongside the slides.

Use the copy or download button to save your work.

Working memory (brief recap)

Working memory has **two key features**: limited capacity, and multiple components.

Attentional control theory uses both features. Only these two are needed here.

Activity 1.1: poll

Optional: Attention, Working memory and executive functions

Fuller treatment of the material here:

- Attention
- Working memory and executive functions

Working memory has limited capacity

Think of working memory as the set of resources for temporarily *holding* and *manipulating* information.

Analogy used often: a working desk. There is enough space for the things you are working on right now, not much else…

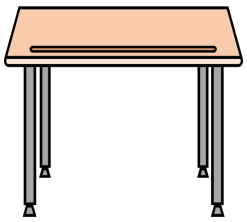


Figure 1: School desk

Desk analogy

The white rectangle is the desk, representing how much working memory is being used at this moment. Currently: unlikely/impossible situation where nothing is taking space in working memory

Desk analogy

Most of the times (or always) something occupies space on the desk: the things that are in the working memory. Sometimes one has enough working memory capacity spare to do well whatever it is that one is doing.

Desk analogy

Other times, there's too much in working memory and the system fails due to overload (running out of cognitive resources). Whatever one is doing is going to suffer in quality.

Working memory's multiple components

Image: Alan Baddeley

Three components: two store information, one manages access to it.

Three components of working memory: central executive (manages access), phonological loop (verbal-phonological storage), visuospatial sketchpad (visuospatial storage).

Anxiety is proposed to disrupt the **central executive**, not the two storage components.

In Session II the central executive is subdivided into three executive functions. Three of the six hypotheses of Attentional Control Theory concern them.

Optional: Development of the working memory model

Alan Baddeley on the development of his working memory model: https://youtube.com/watch?v=mT0NLihOK30

House windows exercise

Helpful to realize how the three components work together.

Take a couple of minutes to **work out how many windows are there in your house**

- You retreive from long-term memory and manipulate a mental image of your house as you "walk through" it room by room (*visuospatial sketchpad*)
- As you progress, you keep a count of the number of windows by rehearsing the numbers out loud or not. (*phonological loop*)
- You will have followed a strategy: deciding which rooms to visualize next, keeping track of progress, and ensuring the count is updated (*central executive*).

Orienting of attention: endo- vs exo-genous

Limited capacity means that not all information can be processed. Selection is required, and that is one function of attention. Orienting can be driven from two directions.

Endogenous / goal-directed

- "endo" = from the inside
- voluntary, intentional
- determined by the person
- top-down
- in sport: *concentration*

Exogenous / stimulus-driven

- "exo" = from the outside

- involuntary, not intentional
- determined by the environment
- bottom-up
- in sport: *distraction*

Examples:

- **Goal-directed**: deliberately turning your head to look where your team-mate will pass
- **Stimulus-driven**: in a tennis tournament, someone in the crowd shouts your name

This distinction underlies the third hypothesis of Attentional Control Theory, stated formally in Session II: anxiety shifts the balance between these two systems.

Attention does other things besides orienting: general alertness, and executive control. Those are covered in the material linked earlier.

"Cocktail party" scenario

Designed to illustrate selective feature of attention at the pre-conscious level, but here merely a shortcut for me to illustrate goal-directed and stimulus-driven attention at play.



Figure 2: Cocktail party

You are in a noisy place. You choose to attend to one speaker (**goal-directed**). Suddenly, your attention is captured (distracted) by a salient, personally relevant cue (like hearing your name from your back) (**stimulus-driven**).

Important: the extent of **attentional control** can be defined as the **balance** between goal-directed (concentration) and stimulus-driven (distraction) attention

What does anxiety do to attentional control?



Figure 3: a meme about anxiety from this module. no actual content

Anxiety types:

- Trait (dispositional)
- State (situational)

Anxiety manifestations:

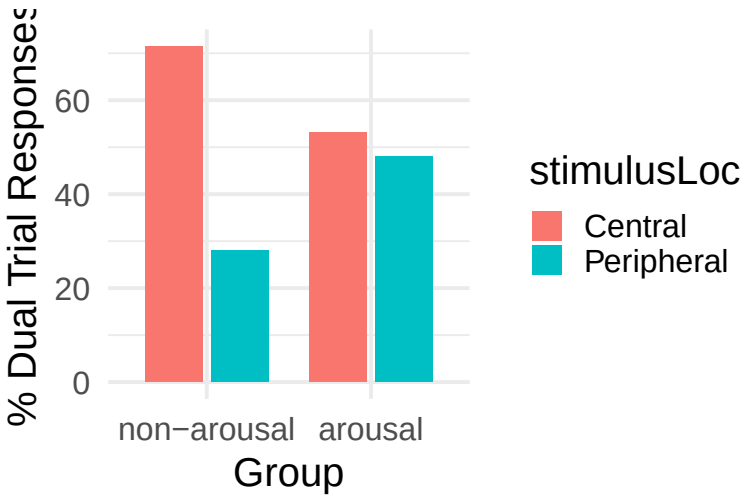
- Somatic
- Behavioural
- Emotional
- Cognitive (worry)

Arousal removes the dominance of central stimuli (Shapiro & Johnson, 1987)

Study 1, key methodology

- N = 30 University students
- **Task**: Classify central and peripheral target stimuli (perceptually difficult because the target stimuli were presented for a short duration.) Using one hand.
- Participants needed to keep their eyes on a central fixation cross. A target stimulus could appear either centrally or in the periphery … or both: dual trial (i.e., two targets at the same time)
- **Groups**: arousal, non-arousal
- The other hand was for receiving *electrical shocks* every now and then, but only the lucky ones in the arousal group.
- Analyses: the researchers were interested in the participants' choices in the dual trials.

Study 1, main results



For the non-arousal group, dominance of central stimulus

For the arousal group, no dominance. Arousal countered the dominance of the central stimulus.

What it shows for ACT: under arousal, task-irrelevant peripheral stimuli compete successfully with the central target: stimulus-driven orienting gains ground.

With the centre less predictable, arousal reverses the dominance (Shapiro & Johnson, 1987)

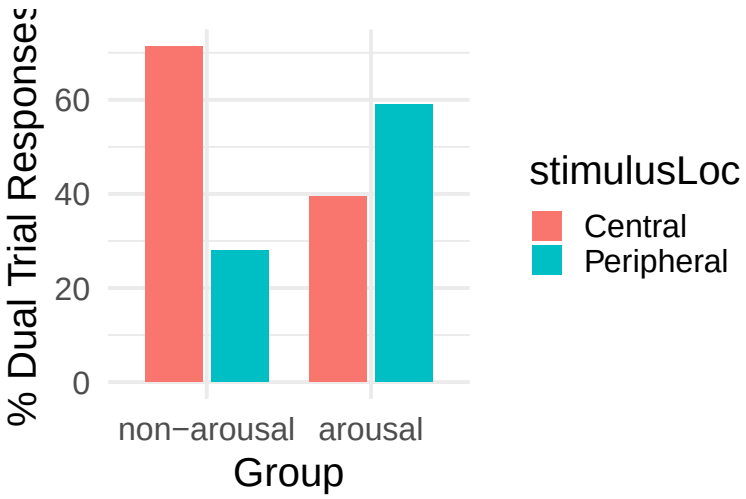
Study 2, key idea

They realized the central stimulus was more predictable (only one location) than the peripheral stimulus. In study 2, they used the same Method, but made the central location less predictable than in study 1.

What do you think they found?

Activity 1.2: form

Study 2, main results



Arousal reversed the dominance, so that when a pair of central-peripheral targets were presented, *the peripheral target became the dominant*.

What it shows for ACT: the shift is not a fixed bias towards the periphery. Arousal favours whichever stimulus is more salient.

Shapiro, K. L., & Johnson, T. L. (1987). Effects of arousal on attention to central and peripheral visual stimuli. Acta Psychologica, 66(2), 157-172. https://doi.org/10.1016/0001-6918(87)90031-X. Accessible from Talis

Threat cues speed detection, more so in phobics (Lipp & Derakshan, 2005)

Key methodology:

- N = 70 participants
- Groups:
  - Spider/snake phobics
  - non-phobics
- Task: emotional dot probe
- Emotional Dot probe: a cue appears on the screen either left or right before a target stimulus appears either left or right. The cue could be neutral or emotional (image of a snake).

Main results:

- The target was detected faster when appearing shortly after an emotional cue
- Larger effects for phobics

What it shows for ACT: threat-related cues capture attention, and what constitutes a threat is specific to the individual.

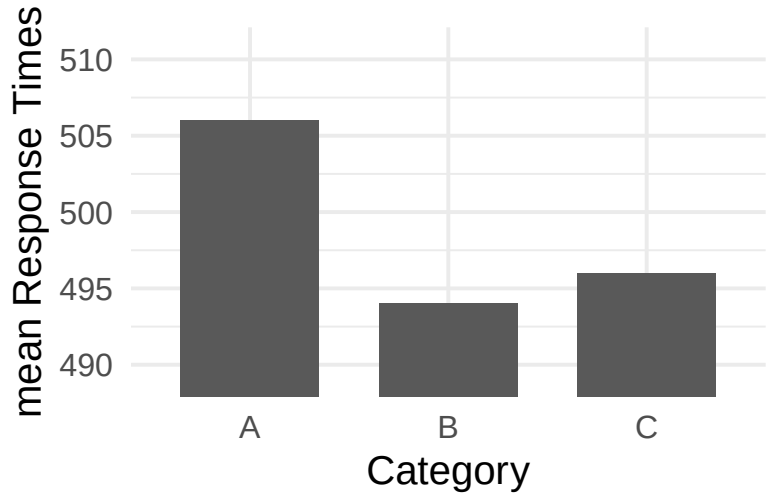
Lipp, O. V., & Derakshan, N. (2005). Attentional bias to pictures of fear-relevant animals in a dot probe task. Emotion, 5(3), 365. https://doi.org/10.1037/1528-3542.5.3.365. Accessible from Talis

Negative sport-related words slow responses in athletes (Lautenbach et al., 2016)

Key methodology:

- N = 40 athletes
- Task: emotional Stroop task
  - A: negative sport related words (e.g., defeat, failure, rival)
  - B: neutral sport related words (e.g., endurance, position, sideline)
  - C: neutral non-sport related words (e.g., radio, lemon, computer)

Main results:



Significantly slower when responding to negative sport-related words.

What it shows for ACT: the interference is specific to personally relevant threat content, in a sport sample, not to negative words in general.

Lautenbach, F., Laborde, S. J. P., Putman, P., Angelidis, A., & Raab, M. (2016). Attentional distraction by negative sports words in athletes under low-and high-pressure conditions: Evidence from the sport emotional Stroop task. Sport, Exercise, and Performance Psychology, 5(4), 296. https://doi.org/10.1037/spy0000073 Accessible from Talis

Summary of effects of anxiety on attention: Attentional Bias

Anxiety induces an **attentional bias**, consisting of enhanced processing of threats:

- *early detection* of threats
- *impaired disengagement* from threats

(whatever threat means to the individual).

Good or bad? Neither. From an evolutionary perspective, anxiety is an adaptive process that increase chances of survival. But thinking about threats might not always be the most functional way to use one's working memory resources.

Worry and an easy task

Resources dedicated to worry (blue) are not dedicated to process task-related information. But there are plenty resources for both. Performance is good.

Greater worry, sufficient resources

If worry takes more resources (i.e., one experiences greater cognitive anxiety), while still leaving enough for the easy task, there is still space for both. Performance is still not impaired.

A more demanding task, insufficient resources

But if the *task is more difficult* requiring more resources, and if there are none left, performance will be impaired.

Performance is impared if there are not enough resources to carry out the task. Worry is a *distraction* that takes resources away from the task

Effort maintains performance at the cost of efficiency

Capacity can be temporarily increased through **effort** mobilisation. Resources are then sufficient for both, and performance is maintained.

Extra resources were required for the same output: **decreased efficiency**.

The same task performed more efficiently

In fact, even a task so difficult could have been carried out well in a more efficient manner (i.e., without the extra effort) if worry did not occupy so much of working memory.

Efficiency as a ratio

The effects of worry (cognitive anxiety) are best studied not on performance, but on *processing efficiency*.

> **Note**
>
> $$\text{efficiency} \propto \frac{\text{performance}}{\text{resources}}$$
>
> The formula is a short version of "how much you do in relation with how much effort you put tells about efficiency". The symbol $\propto$ means "is proportional to".

PET vs ACT

Processing Efficiency Theory (Eysenck & Calvo, 1992), the predecessor of Attentional Control Theory.

PET specifies the cost but not the mechanism, which is what Attentional Control Theory adds.

Eysenck, M. W., & Calvo, M. G. (1992). Anxiety and performance: The processing efficiency theory. Cognition & Emotion, 6(6), 409-434. https://doi.org/10.1080/02699939208409696. Accessible from Talis.

Effort rises with set criticality; performance falls for high-trait-anxiety players (Smith et al., 2001)

Key methodology:

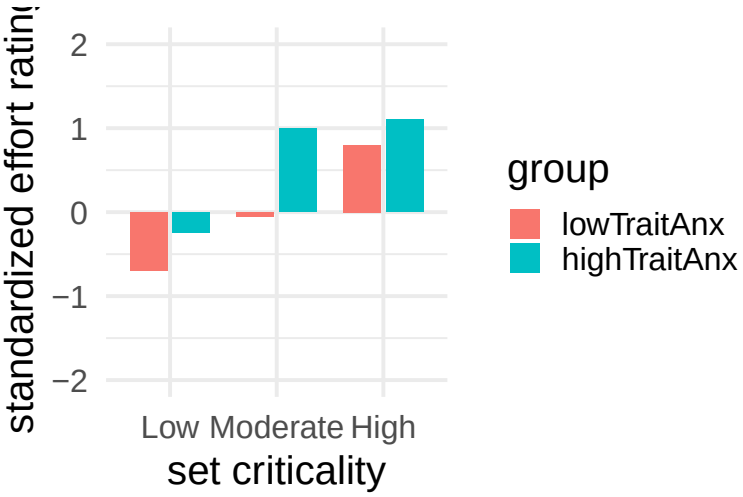
- N = 12 volleyball players
  - 6 low and 6 high in trait anxiety
- Task: Eigth volleyball games
- Measures:
  - State anxiety (through questionnaire)
  - Effort mobilized (through questionnaire)
  - Performance (through video analyses)
- Analysis factors
  - Group (low, high trait anxiety)
  - set criticality
    - low: 7 or more points of difference
    - moderate: between 3 and 6 points of difference
    - high: less than 2 points of difference

Predictions:

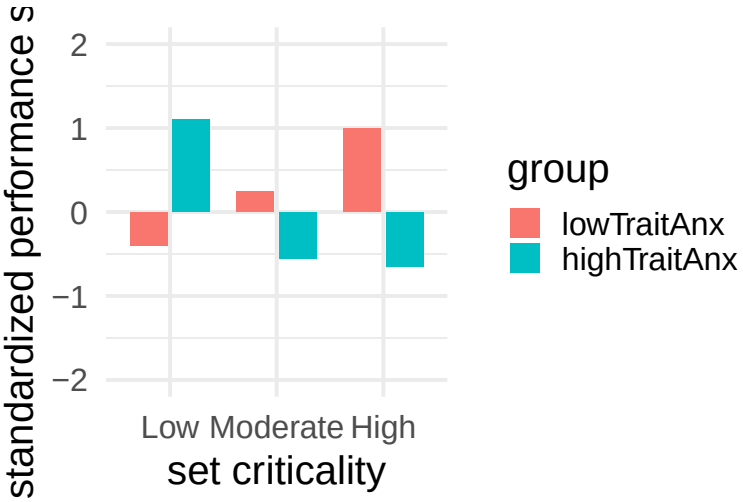
- State anxiety would be greater for more critical sets
- Effort would also be greater for more critical sets
- Performance would increase when more effort is mobilized
  - but only for the low-trait anxiety group

Main results:

- High-trait anxiety groups experienced greater state anxiety



- Effort increased with set criticality



- Performance improved with set criticality for the low-trait anxiety group
- Performance declined with set criticality for the high-trait anxiety group

Interpretation: Performance and worry competed for resources. As worry (state anxiety) increased over set criticality, fewer and

fewer resources were left to focus on the volleyball task. Some support for Processing Efficiency Theory

What it shows for ACT: effectiveness and efficiency dissociate. Effort increased in both groups; only the low-trait-anxiety group converted it into performance.

Smith, N. C., Bellamy, M., Collins, D. J., & Newell, D. (2001). A test of processing efficiency theory in a team sport context. Journal of Sports Sciences, 19(5), 321-332. https://doi.org/10.1080/02640410152006090. Accessible from Talis

Visuospatial attention in target sports: the quiet eye



Figure 4: photo of Joan Vickers

Joan Vickers

As a *variable*

- a duration, in (milli)seconds

- eyes on the critical location or object (e.g., the ball in golf putting)
- begins before movement initiation ("QE onset")
- ends when gaze deviates from that location ("QE offset")

As a *phenomenon*, extensively reported:

- experts show a longer QE than novices
- QE is longer for successful than unsuccessful performance
- it is trainable

Relevance here. On an attentional-control account the Quiet Eye is an index of which system is driving orienting: a long, steady fixation on the task-relevant target indicates *goal-directed* dominance, a short or erratic one indicates *stimulus-driven* dominance.

On this account the quiet eye provides measurable evidence for Attentional Control Theory, and quiet eye training becomes an attentional-control intervention.

The mechanisms underlying the Quiet Eye remains debated.

Quiet eye durations are longer before holed putts (Wilson & Pearcy, 2009)

Key methodology:

- N = 6 participants
- Task: Golf putting on flat and sloped surfaces
- Measure:
  - Performance
  - Quiet Eye duration via eye tracking

Key result:

Optional reading:

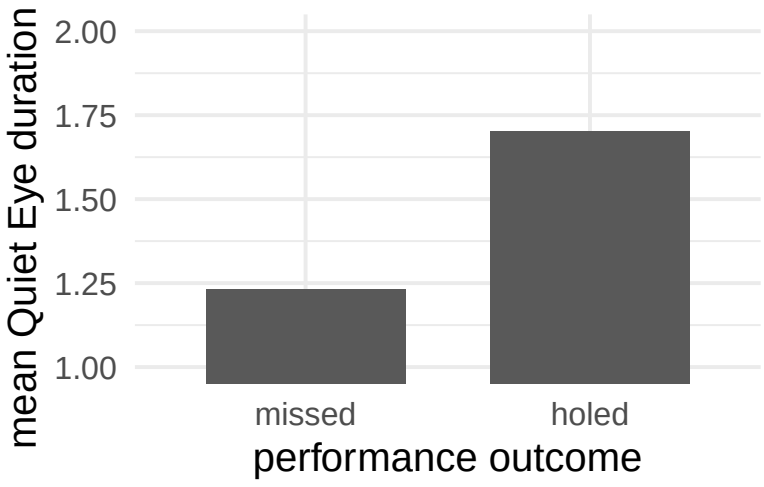
Vickers, J. N. (1992). Gaze Control in Putting. Perception, 21(1), 117-132. https://doi.org/10.1068/p210117 Accessible from Talis

Hossner (Ed.), E.-J. (2016). Quiet Eye research - Joan Vickers on target. Current Issues in Sport Science (CISS), 1, 100. https://doi.org/10.15203/CISS_2016.100

Gallicchio, G., Cooke, A., & Ring, C. (2018). Assessing ocular activity during performance of motor skills using electrooculography. Psychophysiology, 55(7), https://doi.org/10.1111/psyp.13070

Gallicchio, G., & Ring, C. (2020). The quiet eye effect: A test of the visual and postural-kinematic hypotheses. Sport, Exercise, and Performance Psychology, 9(1), 143-159. http://dx.doi.org/10.1037/spy0000162

Quiet Eye on Scientific American Frontiers: https://youtube.com/watch?v=knfC978EoWc



Significantly longer QE durations ahead of holed than missed putts.

What it shows for ACT: within the same performer, successful trials show longer goal-directed fixation. Quiet eye duration varies with control, not with skill level alone.

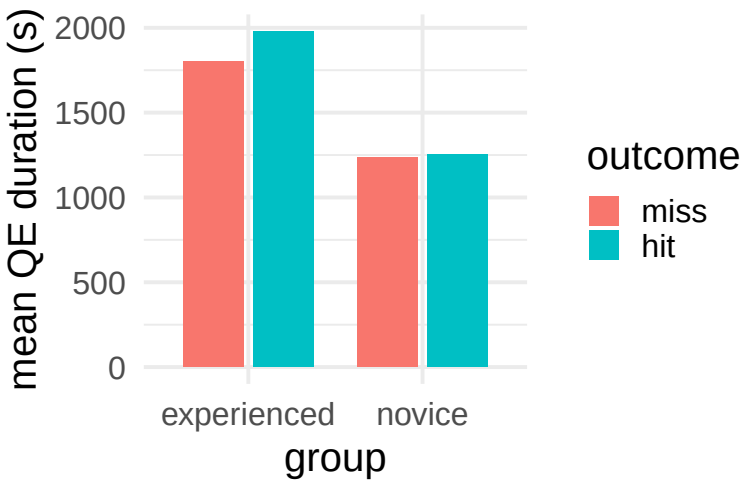
Experienced golfers show longer quiet eye durations than novices (Walters-Symons, Wilson, & Vine, 2017)

Wilson, M. R., & Pearcy, R. C. (2009). Visuomotor Control of Straight and Breaking Golf Putts. Perceptual and Motor Skills, 109(2), 555-562. https://doi.org/10.2466/pms.109.2.555-562 Accessible from Talis

Key methodology

- N = 18 experienced golfers
- N = 21 novice golfers
- Task: flat-surface golf putting

Key result



- Experienced golfers showed longer QE durations than novices

What it shows for ACT: the expert-novice difference supports quiet eye duration as a marker of attentional control.

Effort rises under competition; performance still declines (Wilson et al., 2006)

Walters-Symons, R. M., Wilson, M. R., & Vine, S. J. (2017). The quiet eye supports error recovery in golf putting. Psychology of Sport and Exercise, 31, 21-27. https://doi.org/10.1016/j.psychsport.2017.03.012 Accessible from Talis

Key methodology:

- N = 24 university students
  - Median split into low and high trait anxiety
- Task: simulated rally driving, under two conditions
  - practice (designed to induce lower state anxiety)
  - competition (designed to induce higher state anxiety)
- Measures:
  - Trait and state anxiety (through questionnaires)
  - Performance (time to complete the race)
  - Effort mobilized
    - through questionnaire

    - through pupillometry (increased pupil diameter = more effort)

Main results:

- for both groups, state anxiety increased from practice to competition.
- the high-trait anxiety group showed greater state anxiety.
- Pupil diameter increased from practice to competition for both groups, being also larger for high-trait anxiety group
- Self-report effort showed similar results, with the practice-to-competition increase much larger in the high-trait anxiety group.
- No group difference in performance during practice
- Performance decreased from practice to competition
- Performance drop was larger for the high-trait anxiety group

Interpretation:

- There were enough spare resources in working memory to complete the driving task when worry (state anxiety) used a small amount of resources.
- Worry (state anxiety) occupied more resources in competition, especially for the high-trait anxiety group.
- The high-trait anxiety group mobilized more resources, increasing their working memory, but the extra resources were not enough to maintain as good performance.
- These findings support Processing Efficiency Theory

What it shows for ACT: pupil diameter and self-reported effort both increased, yet performance declined further in the high-trait-anxiety group: the additional resources mobilised were insufficient.

Wilson, M., Smith, N. C., Chattington, M., Ford, M., & Marple-Horvat, D. E. (2006). The role of effort in moderating the anxiety–performance relationship: Testing the prediction of processing efficiency theory in simulated rally driving. Journal of sports sciences, 24(11), 1223-1233.https://doi.org/10.1080/02640410500497667. Accessible from Talis

Symptoms of a diagnosis of impaired attentional control

Anxiety consumes working memory and transfers attention from goal-directed to stimulus-driven control; performers compensate with extra effort, so efficiency drops before performance does.

The presence of several symptoms makes an attentional-control diagnosis plausible.

Symptom	Observable example
Attention captured by task-irrelevant cues under pressure	Report of noticing cameras or spectators; gaze leaving the target late in the routine
Worry reported before or during execution	Report of thinking about the consequences of failure
Effort increased, performance unchanged or reduced	Report of greater training effort without corresponding improvement; visible straining
Difficulty disengaging after an error	One error followed by a cluster of errors
Erratic gaze, or shortened final fixation, in aiming tasks	Measured by eye tracking, or inferred from a rushed pre-performance routine

The same symptoms can be compatible with a different diagnosis or other accounts of performance under pressure.

Learning outcomes, Session II

Six hypotheses, then two interventions and the evidence for them.

By the end of this session you should be able to:

1. Name the three executive functions and relate them to the central executive
2. State attentional control theory's **six** hypotheses
3. Describe research testing the theory, and say what each study shows

4. Define gaze training and cognitive training, and the mechanism targeted by each of them
5. Discuss the evidence for each intervention, including its weaknesses
6. Apply the theory to an athlete and justify an intervention

Recap of session I

Three questions on last session's material.

Activity 2.1: poll

Worksheet 2

Worksheet 2

Space to write during this session: predictions, reasoning, notes.

Open it now and keep it alongside the slides.

Use the copy or download button to save your work.

Brief recap of Processing Efficiency Theory

People who experience greater cognitive anxiety (worry) *might* perform worse than those who experience less. But this is not always the case… why?

- There is not enough space in the working memory to process everything
- Our attention selects information to be processed
- Working memory and attention are not unitary concepts
- **Cognitive anxiety (worry) does not necessarily reduce performance**
- Effort can counteract the impairment **at the expense of processing efficiency**
- Cognitive anxiety (worry) reduces processing efficiency

Executive functions

Central executive → inhibition, shifting, updating

Inhibition = resisting the processing of task-irrelevant information (resisting distractions)

Shifting = moving focus towards task-relevant information (concentrating)

Updating = monitoring and processing information held in working memory

Hypotheses 4, 5 and 6 each state that anxiety impairs one of these functions.

This greater specificity about the central executive is what developed Processing Efficiency Theory into Attentional Control Theory.

Attentional Control Theory: six hypotheses

Building on the claims stated in Session I, here are the **six hypotheses** that define Attentional Control Theory.

ACT's six hypotheses: 1. processing efficiency, 2. task demands, 3. attentional control, 4. inhibition, 5. shifting, 6. updating

Hypotheses 1 and 2 concern the cost of anxiety. Hypotheses 3 and 4-6 concern the mechanism.

Optional

Michael Eysenck on the bi-directional effects of anxiety and cognition: https://youtube.com/watch?v=7j47tjYzeKY

Eysenck, M. W., Derakshan, N., Santos, R., & Calvo, M. G. (2007). Anxiety and cognitive performance: attentional control theory. Emotion, 7(2), 336. https://doi.org/10.1037/1528-3542.7.2.336 Accessible from Talis

ACT hypothesis 1: processing efficiency

Cognitive anxiety impairs processing *efficiency* more than it impairs performance.

Worry (in blue) occupies resources in Working Memory. But there are enough resources to complete the task (in yellow).

ACT hypothesis 1: processing efficiency

Cognitive anxiety impairs processing *efficiency* more than it impairs performance.

Worry (in blue) takes resources away from the task (in yellow). Performance is impaired.

ACT hypothesis 1: processing efficiency

Cognitive anxiety impairs processing *efficiency* more than it impairs performance.

Overall resources (in white) are temporarily increased through **greater effort**. Performance of the task (yellow) is not impaired despite the large worry (blue). **Efficiency decreases**.

ACT hypothesis 2: task demands

The more difficult the task, the higher the chances of decreased efficiency

If the task is **easy** (i.e., it requires not many Working Memory resources), the effects of worry on efficiency are not visible (no need to put extra effort).

ACT hypothesis 2: task demands

The more difficult the task, the higher the chances of decreased efficiency

It is when the task is **hard** (i.e., it requires many Working Memory resources), that the effects of worry on efficiency are more likely to be visible (there is need to put extra effort).

ACT hypothesis 3: attentional control

Cognitive anxiety disrupts attentional control

A scale analogy:

goal-directed (i.e., concentration) and stimulus-driven (i.e., distraction) types of orienting compete for which one is the main driver of attention/selection (i.e., what to focus on)

Attentional control = ability to maintain a goal-directed (endogenous) attentional orienting (concentration), while suppressing stimulus-driven (exogenous) attentional orienting (distraction)

The issue when performing under pressure:

GOAL-DIRECTED STIMULUS-DRIVEN

You lose focus on the task and become distracted by non-task related information.

The objective of performing under pressure intervention:

GOAL-DIRECTED STIMULUS-DRIVEN

You shield from distractions and maintain focus on task-related information.

ACT hypothesis 4: inhibition

How is attentional control disrupted (hypothesis 3) under elevated cognitive anxiety? Through **impaired inhibition**

- Impaired inhibition enhances the stimulus-driven path to attentional orienting
- One becomes unable to block out irrelevant information
- Potential distractions (non-task related activity) intrude in one's attention

ACT hypothesis 5: shifting

How is attentional control disrupted (hypothesis 3) under elevated cognitive anxiety? Through **impaired shifting**

- Impaired shifting decreases the goal-directed path to attentional orienting
- One becomes unable to focus on the relevant information
- Potential important information (task related activity) are not in one's focus

ACT hypothesis 6: updating

How is attentional control disrupted (hypothesis 3) under elevated cognitive anxiety? Through **impaired updating**

- When the Working Memory is overloaded, it loses the ability to update and monitor information
- One fails to keep track of changing task demands
- Processing becomes less flexible and adaptive

Pressure shortens quiet eye duration (Wilson et al., 2009)

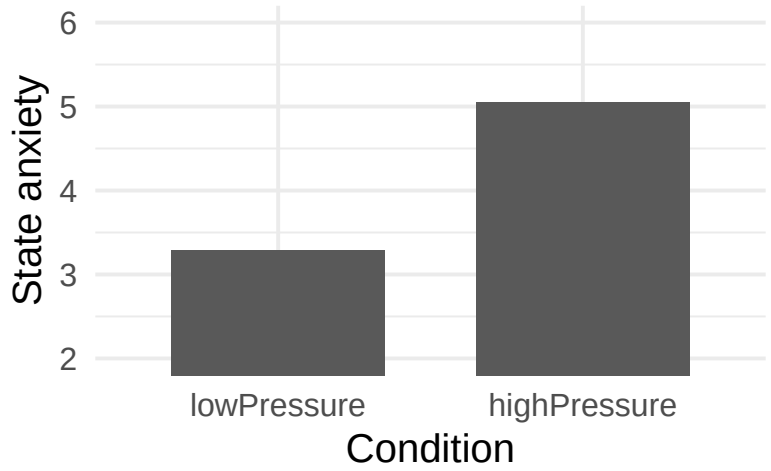
Key idea:

- Quiet Eye duration as measure of attentional control
- Does anxiety impair QE durations?

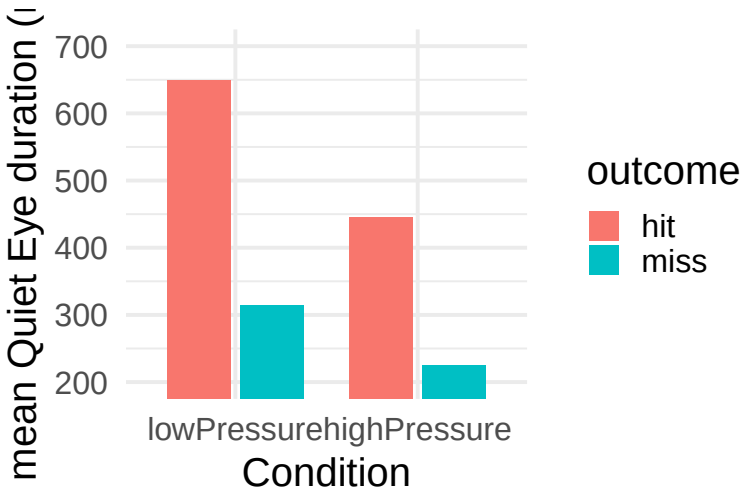
Key methodology:

- N = 10 university students (basketball players)
- Task: free throws until 10 hits and 10 misses
- Conditions:
  - Control (low pressure)
  - Experimental (high pressure, competition and financial incentive)
- Measures:
  - State anxiety, through questionnaire
  - Quiet Eye duration, through eye tracking

Key results:



State anxiety was larger in the experimental condition.



Effect of outcome: Shorter Quiet Eye duration for missed throws.

Effect of pressure: Shorter Quiet Eye duration under greater pressure.

Some evidence that attentional control is impaired under elevated state anxiety

What it shows for ACT: direct evidence that anxiety impairs attentional control in a sport task, measured rather than inferred.

Wilson, M. R., Vine, S. J., & Wood, G. (2009). The influence of anxiety on visual attentional control in basketball free throw shooting. Journal of Sport and Exercise Psychology, 31(2), 152-168. https://doi.org/10.1123/jsep.31.2.152 Accessible from Talis

Interventions

Gaze training targets hypothesis 3 (attentional control). **Cognitive training** targets hypotheses 1 and 4 (efficiency, inhibition).

Attentional control can be addressed in two ways: by training **gaze behaviour** (as proxy for attentional control), or by training **working memory capacity**.

For each: what it involves, the ACT-specific mechanism it addresses, and the supporting evidence.

The evidence is not equally strong for both. One is well supported (Quiet Eye); the other is promising but less tested (working memory training).

Gaze training / Quiet Eye training

Definition: training to produce a longer and steadier quiet eye, specifically under elevated cognitive anxiety.

ACT-specific mechanism addressed (hypothesis 3):

- Quiet eye duration indexes the dominance of goal-directed orienting (concentration) over stimulus-driven orienting (distraction)
- Training a longer quiet eye trains the dominance of goal-directed orienting

Delivery. The intervention consists of instructions: a fixed pre-performance gaze routine, delivered verbally, rehearsed in practice and then under induced pressure. Eye tracking is required to measure the quiet eye and to sanity-check the intervention, but not to deliver the training.

Caveats to consider for the success of training:

- most experts already show long quiet eye durations, at least under low pressure
- in many sports it is not straightforward to identify where the visual focus should be, or for how long

Quiet eye training maintains putting performance under pressure (Vine & Wilson, 2010)

First Quiet Eye training study

Key methodology:

- N = 14 university students, minimal golf experience

  - control goup: technical training
  - experimental group: technical and Quiet Eye training

- Design: longitudinal (i.e., over time)
- Task: straight putts

Procedure:

Study procedure:

1. Pre-test
2. Training (8 sessions): Control vs Experimental
3. Post-test: Retention 1 → Transfer (high pressure) → Retention 2

Training protocol instructions

Technical instructions:

- Stand with legs hip width apart and keep your head still.
- Maintain relaxation of shoulders and arms.
- Keep the putter head square to the ball.
- Perform a pendulum-like swing and accelerate through the ball.
- Maintain a still head after contact.

Quiet Eye training instructions

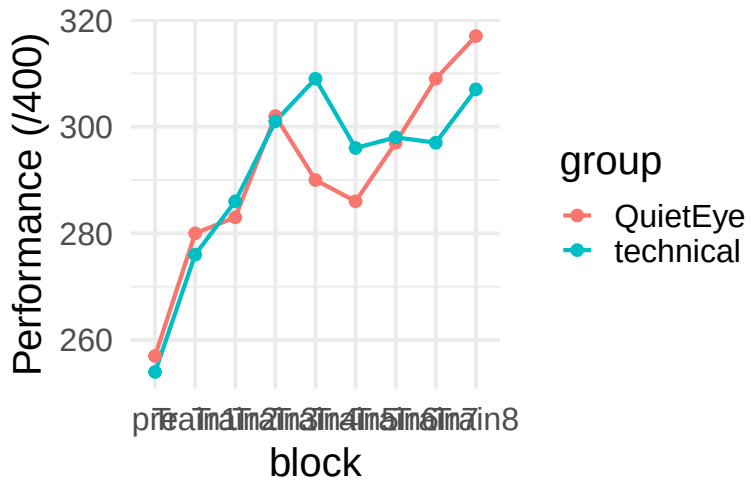
- Assume stance and ensure that gaze is on the back of the ball.
- Fixate the hole. (Fixation should be made no more than three times.)
- Your final fixation should be on the back of the ball and for not longer than 2-3 seconds.
- No gaze should be directed to the club head or shaft during the putting action.
- Your fixation should remain steady for 200-300 ns after contact with the ball.

Activity 2.2: form

Key results:



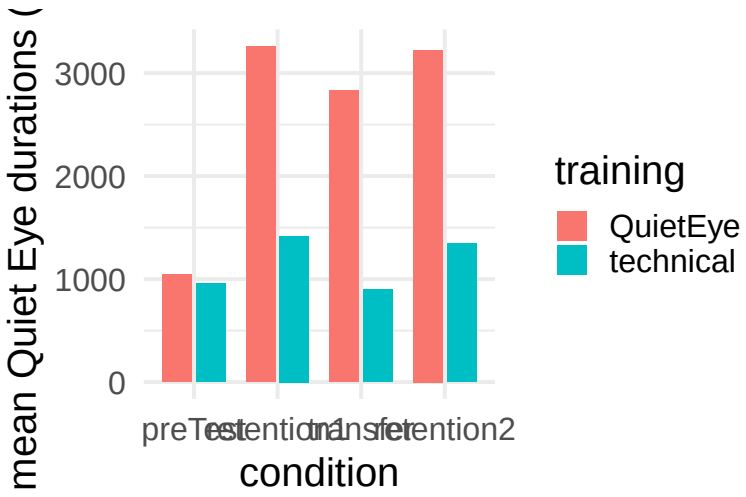
The training did not have an effect on state anxiety. The training did not alter people's anxiety reaction to competitive pressure.



No performance differences at pre-test (before training).

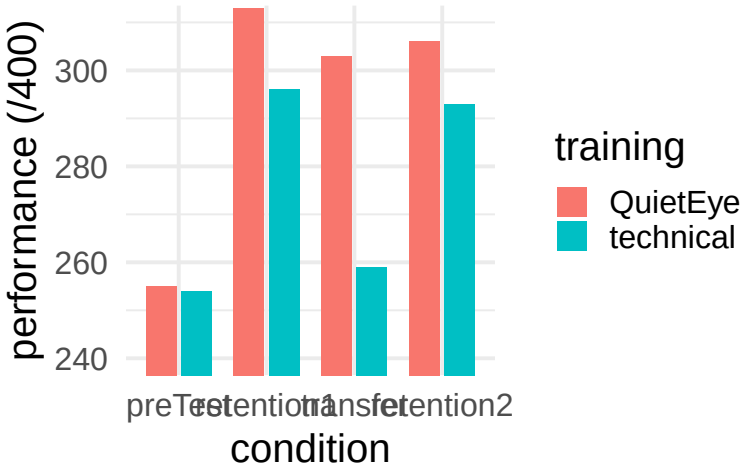
Performance overall improving for both groups.

No obvious performance advantage on one group over the other.



Quiet Eye durations were as expected:

- equal between groups at pre-test
- longer for the QE group, at the post-tests



Performance was comparable across groups at all retention post-tests, except at *transfer* (high pressure performance), where the technical but not the Quiet Eye group showed a significant drop in performance

What it shows for ACT: gaze training altered the effect of pressure on performance without altering reported anxiety. The

mechanism operated on attention rather than on emotion.

Vine, S. J., & Wilson, M. R. (2010). Quiet eye training: Effects on learning and performance under pressure. Journal of Applied Sport Psychology, 22(4), 361-376. http://dx.doi.org/10.1080/10413200.2010.495106 Accessible from Talis

Optional reading (review of Quiet Eye training literature): Vine, S. J., Moore, L. J., & Wilson, M. R. (2014). Quiet eye training: The acquisition, refinement and resilient performance of targeting skills. European journal of sport science, 14(sup1), S235-S242. http://dx.doi.org/10.1080/17461391.2012.683815 Accessible from Talis

Cognitive training

Definition: training to increase working memory capacity.

ACT-specific mechanisms addressed (hypotheses 1 and 4-6):

- worry occupies resources that would otherwise be available for task-related processing
- a permanently larger capacity would leave sufficient resources for the task despite worry
- improved inhibition would reduce the intrusion of task-irrelevant information

Delivery. Repeated sessions of an adaptive working memory task; in the studies below, computerised dual n-back over approximately three weeks.

Caveats to consider for the success of training:

- transferability to real-world tasks is disputed
- the sport evidence comes mainly from a single research group

Adaptive dual n-back improves attentional control under anxiety (Sari et al., 2016)

Key methodology:

- N = 26 university students, pre-screened through questionnaires for high trait anxiety and low attentional control
  - experimental group: adaptive cognitive trainining (adaptive dual n-back)
  - control group: low-load cognitive training (non-adaptive dual 1-back)
- Design: longitudinal (i.e., over time)

Procedure:

Study procedure:

1. Pre-test
2. Training (15 sessions): Control group vs Experimental group
3. Post-test

- Some *pre* and *post*-test measures:
  - attentional control through rest state electroencephalography (EEG)
  - Flanker task performance (Link to Flanker task), under anxiety-inducing conditions (loud and unpredictable noise bursts)

Training explanations:

- visual n-back: one 3x3 grid is presented at the time with only once cell illuminated. Participants need to respon when the same grid was presented n trials before.
- phonological n-back: one letter is pronounced at the time. Participants need to respon when the same letter was presented n trials before. (Link to classic visual letter n-back)
- dual n-back: visual and phonological stimuli are presented simultaneously. Trials can match visually (press a key), phonologically (press another key), or both (press both keys)
- adaptive dual n-back: the number n increased or decreased at the next block depending on accuracy at the previous block. Consult the paper for a figure of the stimuli
- Schedule: 15 sessions in 21 days

Some key results

- the adaptive group reached high levels of complexity (large n in b-back)
- The adaptive group showed better Flanker task performance under conditions of elevated state anxiety
- The adaptive group showed larger values of EEG indices associated with better attentional control

- Cognitive training improved attentional control under pressure / resilience of attentional control to pressure

What it shows for ACT: working memory capacity can be trained, with the benefit appearing specifically under elevated anxiety. This is the laboratory evidence.

Adaptive dual n-back in a tennis task (Ducrocq et al., 2017)

Sari, B. A., Koster, E. H., Pourtois, G., & Derakshan, N. (2016). Training working memory to improve attentional control in anxiety: A proof-of-principle study using behavioral and electrophysiological measures. Biological psychology, 121, 203-212. http://dx.doi.org/10.1016/j.biopsycho.2015.09.008 Accessible from Talis

Key methodology:

- Similar to that of Sari et al. (2016) but in a sport task (tennis)
- N = 30 participants from tennis club
  - experimental group: adaptive dual n-back
  - control group: non-adaptive dual 1-back
- Some pre- and post-test measures:
  - performance at change detection task
  - tennis volley task performance and Quiet Eye duration, under lower and higher pressure

Some key results

- The experimental group reached performed better than the control group at the change detection task
- The experimetal group performed better at the higher pressure, whereas no changes in the control group
- The experiemntal group showed later Quiet Eye period offset than the control group

What it shows for ACT: the sport evidence for the same claims.

Design considerations. Approximately fifteen participants per group; no pre-training pressure test; a fixed low-then-high pressure order.

Ducrocq, E., Wilson, M., Smith, T. J., & Derakshan, N. (2017). Adaptive working memory training reduces the negative impact of anxiety on competitive motor performance. Journal of Sport and Exercise Psychology, 39(6), 412-422. https://doi.org/10.1123/jsep.2017-0217 Accessible from Talis

Effectiveness and mechanism are separate questions

Evidence that quiet eye training is **effective** is separate from evidence about the **mechanism** by which it works.

The first is reasonably strong: trained performers maintain performance under pressure better than technically trained controls.

The second is debated.

An intervention can be described as effective and interpreted through Attentional Control Theory, provided the mechanism is presented as debated.

Imposed hold vs trained quiet eye

A coach instructs an athlete in a target sport (e.g., golf) to hold the sight on the target (e.g., the ball) for longer before release. The athlete concentrates on executing that instruction.

Is this quiet eye training?

No.

A consciously imposed fixation adds an explicit instruction to perform and monitor.

Training quiet eye means rehearsing a gaze routine until (ideally) it becomes automatic, requiring *less* conscious processing under pressure rather than more.

Summary: applying Attentional Control Theory

The components of an argument aligned with Attentional Control Theory.

The theory

Anxiety consumes working memory and transfers attention from goal-directed to stimulus-driven control; performers compensate with extra effort, so efficiency drops before performance does.

Supporting studies

- Shapiro & Johnson (1987) — arousal (anxiety) shifts which stimulus dominates
- Smith et al. (2001) or Wilson et al. (2006) — effort rises, efficiency falls
- Wilson et al. (2009) — pressure impairs attentional control / shortens quiet eye, in turn impairing performance

Observable symptoms

Task-irrelevant cues intruding under pressure; worry reported before execution; effort up with performance flat; difficulty disengaging after an error; gaze becoming erratic.

Two interventions

- Gaze training, targeting hypothesis 3
- Cognitive training, targeting hypotheses 1 and 4-6

Focus on the mechanisms

Identify the mechanism implicated, then show that the intervention acts on that mechanism.

Optional: some thoughts

Which model of performing under pressure is right?

Attentional control

Reinvestment

Challenge and threat appraisal

Psychomotor efficiency

Constrained action

Ironic processing

Etc.

No doubts these models will be refined again and again, and that's the researcher's job. For applied contexts, however, I encourage you to focus on the *utility* of a model (in explaining a phenomenon or achieving practical impact).

Image: George E. P. Box

George E. P. Box Read more.

> "All models are wrong, but some are *useful*.